

Innovative startups and their traditional peers: Further evidence using performance and survival analysis

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Abstract

This study assesses the Italian Startup Act, introduced in 2012, to promote innovative entrepreneurship. Focusing on manufacturing firms registered between 2014 and 2017, we compare innovative startups with non-innovative peers using matching techniques, survival analysis, probit models, and panel regressions. Findings reveal that innovative startups exhibit stronger financial structures, higher survival rates, and fewer credit constraints, though they take longer to achieve profitability. Firms with greater leverage are more likely to register as innovative, while larger firms are less inclined to do so. Intellectual property (IP) compliance correlates with higher survival and stronger patent activity. These results offer insights into industrial policy effectiveness. While the Startup Act supports financial access and firm resilience, its potential to drive innovation and IP development remains underexploited. Our analysis suggests the need for more targeted, regionally nuanced interventions to enhance the policy's impact and better support the growth of Italy's innovation-driven startup ecosystem.

KEYWORDS

high-tech, innovation, intellectual property, Italy, SMEs, startup

JEL CLASSIFICATION

G30, G32, G38, H81, O30

1 | INTRODUCTION

Europe's industrial sector has been navigating complex structural challenges in recent decades—accelerating technological change, fierce global competition, and the growing imperative to align with sustainable and digital transitions. These trends have underscored the strategic importance of innovation and entrepreneurship as engines of economic resilience, productivity, and employment. Within this evolving landscape, industrial policy has re-emerged as a vital tool, particularly in countries like Italy, where persistent gaps in innovation performance, youth employment, and regional development continue to inhibit inclusive growth.

While rich in industrial tradition and home to a dynamic network of small and medium-sized enterprises (SMEs), the Italian economy lags behind its European counterparts in key innovation metrics. According to the European

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Innovation Scoreboard, Italy remains a “moderate innovator,” with below-average performance in R&D investments, digital competencies, and patent activity. These deficiencies include broader structural issues, including limited access to finance, weak university-business linkages, and a persistent brain drain of young graduates seeking better opportunities abroad.

Amid these challenges, promoting innovative startups has gained traction as a policy lever to revitalize Italy's industrial base. Startups are often lauded for their agility, risk-taking behavior, and capacity to generate disruptive technologies. They also play a pivotal role in fostering high-quality employment and catalyzing new forms of human capital that blend technical, managerial, and entrepreneurial skills. Yet, even the most promising ventures struggle to survive and scale without an enabling ecosystem.

Responding to this need, the Italian government introduced the “Startup Act” in 2012—a comprehensive legislative initiative designed to create a more favorable environment for innovative entrepreneurship. The Act introduced a suite of fiscal, administrative, and financial incentives, including simplified incorporation procedures, tax breaks for investors, and preferential access to public guarantees and crowdfunding platforms. However, despite the proliferation of registered innovative startups in the years following the Act's implementation, questions remain about the effectiveness of these measures in addressing the systemic constraints entrepreneurs face.

Much of the existing literature on Italian innovation policy has focused on short-term or descriptive evaluations, often lacking rigorous counterfactual analysis or sector-specific insights. Furthermore, while a number of studies have examined startup outcomes in aggregate terms, few have disentangled the distinct trajectories of innovative versus non-innovative firms, particularly within manufacturing—a sector critical to Italy's industrial competitiveness. This gap leaves a crucial policy question unanswered: to what extent has the Startup Act succeeded in fostering financially resilient, technologically advanced, and sustainably growing startups?

This study seeks to fill that void by conducting a comprehensive, longitudinal evaluation of the Italian Startup Act's effectiveness. Using a matched sample of manufacturing firms registered between 2014 and 2017, we assess the differential impact of the policy on performance, survival, and innovation outcomes among innovative and traditional startups. Specifically, we explore the following research questions: (1) What are innovative startups' defining features and performance patterns compared to their non-innovative counterparts? (2) How do statutory incentives and legal requirements influence access to finance, growth, and survival? (3) To what degree does intellectual property activity serve as a lever for innovation and sustainability? And (4) What lessons can be drawn to refine the current policy framework and foster a more inclusive and impactful innovation ecosystem?

By addressing these questions, this paper contributes a nuanced understanding of the structural and financial dynamics shaping Italy's startup landscape. It also offers evidence-based insights for designing more targeted and effective industrial policies that align innovation objectives with broader economic development goals.

The structure of our study is organized as follows: the next section introduces the Italian framework for innovative startups, followed by an analysis of relevant literature on startups and innovation. Subsequent sections outline the objectives and methodological approach of the study, present the findings, and discuss avenues for future research.

2 | THE POLICY FRAMEWORK FOR INNOVATIVE STARTUPS IN ITALY

The Italian government implemented the “Startup Act” through Decree-Law n to foster innovation and job creation. 179¹ and Law n. 221 in 2012. This legislative package introduced a comprehensive suite of measures aimed at enhancing the startup ecosystem, including (1) Reduction of bureaucratic hurdles, enabling digital and cost-free incorporation of startups; (2) Implementation of simplified exit procedures, ensuring a more secure process for all stakeholders involved, particularly facilitating the insolvency procedure²; (3) Alleviation of financial constraints, notably through a public guarantee scheme for bank credit and tax incentives for equity investors, alongside the promotion of crowdfunding platforms; (4) Enhancement of the capability to attract innovative and highly specialized talent, facilitated by more flexible employment contracts and startup visa programs designed to attract foreign entrepreneurs.

To avail themselves of these benefits, firms must satisfy specific eligibility criteria to be registered in the Register of Innovative Startups, a dedicated section of the Chamber of Commerce Business Register. The primary qualifications include: (1) The firm must be a limited liability entity (inclusive of joint-stock companies, limited liability partnerships, simplified limited liability partnerships, or cooperative limited liability companies); (2) The age of the firm must be under five years, with a production value not exceeding €5 million; (3) The firm must be engaged in a technology-oriented sector; (4) The firm should represent a genuine startup, not derived from a spin-off or merger of pre-existing companies.

Additionally, until a firm is recognized as an innovative startup, it is precluded from distributing dividends or listing on a stock exchange. Given these stipulations, startups are more susceptible to various market failures, necessitating governmental support, particularly within the Italian context, where financial markets exhibit gaps in the intermediation chain (Anderloni & Braga, 2019). Moreover, it is essential to note that the incentive framework mandates incorporating digital technologies in every administrative procedure, a measure aimed at simplifying processes and fostering a digital culture.

The observed growth was predominantly rapid, with a significant concentration in service-related activities, followed by those in the manufacturing sector. Geographically, the distribution of this phenomenon exhibits a marked imbalance, with a pronounced concentration in the northwest region, while other regions demonstrate a relatively equal distribution, excluding the peripheral islands.

3 | RELATED LITERATURE

The discourse on innovation and economic growth is rich and multifaceted, with significant contributions delineating the pivotal role of startups and small to medium-sized enterprises (SMEs) in fostering innovation. These entities are celebrated for their propensity to invest in high-risk, cutting-edge projects, a characteristic less commonly observed in larger, more established firms (Acs & Audretsch, 1988; Baumol, 2004; Kerr & Nanda, 2014). This phenomenon is further substantiated by reports from the European Commission in 2008, 2012, and 2021, which underscore the unique contribution of these firms to innovation landscapes. Particularly noteworthy are “gazelles” – a subset of small and young firms identified for their exceptional growth potential, significant contributions to entrepreneurship culture, and job creation (Acs & Mueller, 2008).

However, the crux of fostering such innovation often hinges on the availability of finance. Berger and Udell's (1998) framework elaborates on the evolutionary financial needs of startups, beginning with internal finance and equity from angel investors, progressing to external debt and public equity as the firms mature and their financing ecosystems develop effectively. This trajectory, however, is fraught with challenges, particularly in the Italian context, where the functionality of financial intermediaries and markets often falls short of supporting entrepreneurial finance (Anderloni, 2019a, 2019b; Anderloni & Braga, 2019).

A salient barrier to financing for innovative startups and SMEs is the prevalence of information asymmetries, leading to adverse selection and moral hazard problems. Factors such as the absence of a credit history, the inherently risky nature of innovative projects, and the difficulties in collateralizing intangible assets compound these challenges, fostering financial rationing (Beck & Demirguc-Kunt, 2006; Bridges & Guariglia, 2008; Hall & Lerner, 2010). The European Investment Fund (EIF, 2020) highlights the critical nature of these financial impediments, noting the starkly low proportion of European startups that scale, attributed to constrained financing opportunities.

The broader economic literature also shows the interplay between financial infrastructure and innovation. Studies have shown that industries reliant on external finance and those more technologically intensive tend to exhibit higher levels of innovation in environments with well-developed equity markets (Hsu et al., 2014). This underscores the significance of a robust financial ecosystem in facilitating technological advancement and economic growth.

The contemporary literature on startup innovation explores multiple dimensions, including spatial externalities, collaborative dynamics, policy impacts, and innovation strategies. Spatial externalities, as assessed in the survival of Italian startups, demonstrate dual impacts—negative during early stages but supportive post-maturity (Giuliani et al., 2023). Cooperation and technological orientation are pivotal, boosting startups' capacity to innovate and adapt (Cecchele Lago et al., 2023). In Poland, external attractiveness hinges on capabilities, market potential, and clarity of business indicators to secure investments (Jędrzejczyk, 2023). The Lean Startup approach, emphasizing iterative development and customer insights, enhances adaptability and market alignment (Oleka, 2023). Accelerators serve as open innovation hubs, linking startups to knowledge networks that enhance radical and incremental innovations (Del Sarto et al., 2022). However, innovativeness may inversely correlate with survival due to increased risk and resource demand during nascent stages (Hyytinen et al., 2015). Policies such as Italy's Startup Act reflect mixed outcomes, offering fiscal incentives for equity while competing with loan guarantees (Giraudo et al., 2019). Human capital within corporate venture capital units drives both exploratory and exploitative innovation (Bendig et al., 2024). Lastly, crowdfunding platforms like Kickstarter redefine startup success by integrating sustainability with mindful consumption practices, influencing consumer engagement (Gómez-Olmedo et al., 2024).

Recent scholarly contributions specific to the Italian industrial policy and startup ecosystem offer preliminary insights but often suffer from limitations in scope and time horizon (Brancalani et al., 2022; Ferrucci et al., 2021;

Finaldi Russo et al., 2016; Albanese et al., 2019; Giraudo et al., 2019; ISTAT, 2022; OECD, 2018). These studies, while valuable, tend to focus narrowly on either the innovation output or financial activities of firms, lacking a comprehensive view of the interdependencies between these facets.

Our research seeks to bridge this gap by leveraging a more extended period of analysis and focusing on the manufacturing sector, thereby providing nuanced insights into the impacts of industrial policies on innovation and financial performance. We draw on the contrasting findings of previous studies, such as those by Finaldi Russo et al. (2016) and Giraudo et al. (2019), to underscore the differential effects of public policies on manufacturing versus service sectors regarding access to finance and innovation incentives.

Moreover, our examination of the role of intellectual property rights, especially patents, in facilitating access to finance for innovative companies adds a critical dimension to the discourse on innovation financing. By adopting a granular approach to the analysis of patent data, we offer a more refined understanding of how intellectual property contributes to distinguishing innovative firms from their non-innovative counterparts (Hall et al., 2014).

In sum, our study not only corroborates previous research findings but also enriches the literature by providing comprehensive evidence on the positive impacts of Italy's Startup Act on both innovation and financial performance in the startup ecosystem. Through this extended time horizon and focused analysis, we affirm the long-lasting benefits of targeted industrial policies in fostering a conducive environment for the growth and innovation of startups and SMEs.

4 | METHODOLOGY

This study provides a nuanced analysis of the Italian startup ecosystem, evaluating the efficacy of the “Italian Startup Act” in fostering innovation and supporting the growth of innovative startups. Utilizing a robust dataset, we compared the performance of innovative startups against their non-innovative counterparts using rigorous econometric methods, including survival analysis, propensity score matching, and pooled regression models. We also performed a Differences-in-Differences (DiD) analysis; the results are not reported here but are available upon request. The findings offer both insightful outcomes and implications for policy refinement. Moreover, we did not apply winsorization to the variables in this study. Instead, we addressed potential outliers through log transformations and robust regression specification.

4.1 | Data

This study conducts a comprehensive policy evaluation of the Italian “Startup Act,” aiming to derive actionable policy implications that can enhance the efficacy of the policy by amplifying its positive effects and minimizing associated costs. Our focus is on innovative startup firms in Italy's manufacturing sector, registered between 2014 and 2017, examining their performance over five years up to 2021. This analysis utilizes data from the “startup registry” – a specialized section within the official Business Registry of Italy, concentrating exclusively on the manufacturing industry to scrutinize innovation dynamics within this crucial economic sector (see Table 1).

To ensure external validity and an unbiased policy assessment, our research constructs a pooled dataset, incorporating a control group selected from the ORBIS³ database, matched by NACE codes corresponding to innovative manufacturing startups. Financial and governance information was sourced from ORBIS, while intellectual property data was collated from various reliable sources, including ORBIS Intellectual Property. Moreover, the firms were classified as *high-tech* or *low-tech* (a dummy variable differentiating the two groups of firms) according to the European Union's classification, leveraging the NACE-rev. 2 code.

Preliminary data analysis reveals an ascending trend in establishing innovative manufacturing startups spurred by policy incentives. Geographically, the Northern regions (both East and West) emerge as more conducive for innovative startups, followed by the Central, Southern, and Island regions (Table 2-Panel A), reflecting the broader industrialization pattern of the country. Limited liability companies predominate in terms of legal structures, with a minor presence of simplified liability companies and only occasional instances of other forms (Table 2-Panel B). Notably, the analysis of technology intensity within the manufacturing⁴ sub-sectors indicates a greater prevalence of innovative startups in high-tech areas than in their low-tech counterparts, marking a distinct contrast with the pattern observed in the non-innovative business landscape (Table 2-Panel C).

TABLE 1 The innovative startups in Italy by year and sector.

	2013	2014	2015	2016	2017	2018	2019	2020	2021	Total	%
New innovative startups	3	23	306	898	1701	2082	2570	3435	2819	13,837	
Industry											
Agriculture and fishing				2	13	8	30	26	20	99	0.7%
Trade		2	18	29	52	69	69	112	109	460	3.3%
Industry and handcraft	1	3	47	184	306	369	442	548	371	2271	16.4%
Services	2	15	234	673	1297	1614	1994	2711	2294	10,834	78.3%
Tourism		3	5	4	25	14	32	31	16	130	0.9%
Other			2	6	8	8	3	7	9	43	0.3%
Total	3	23	306	898	1701	2082	2570	3435	2819	13,837	100.0%
Region											
North West										4786	34.4%
North East										2778	19.9%
Centre										2905	20.9%
South										2619	18.8%
Island										837	6.0%
Total										13,925	100%

Source: Our analysis of MISE-ISTAT 2018 data.

4.2 | The matching process

To boost the external validity and ensure a more uniform characterization of innovative startups, we employed Propensity Score Matching (PSM). This method leverages observed pre-treatment characteristics of firms to construct a control group with analogous attributes. It is carried out by modeling the probability of participation and matching participants with non-participants based on this probability, utilizing the one-to-one nearest-neighbor technique for precision. Additionally, the matching framework facilitates the application of the Difference in Differences (DiD) methodology and the *t*-test of means. These approaches are instrumental in assessing the impact of the Startup Act by comparing the outcomes for treated and control groups before and after enrollment in the program.

A treatment variable, denoted as *T*, was designated to pinpoint firms participating in the innovative startup program. The adoption of a probit model is justified by its compatibility with the prerequisites of conditional independence and common support, thereby estimating the propensity score for firms to be classified into the treated cohort, where *T* symbolizes the dependent variable (indicative of being an innovative startup), and *X* encompasses the pre-treatment attributes influencing the likelihood of assignment to the treatment group, expressed as:

$$p(X) = \text{prob}(X) = E(T|X) \quad (1)$$

To ensure the integrity of the matching outcome, we conducted year-by-year matches, taking into account various characteristics: (i) NACE activity code, (ii) legal form, and (iii) geographical location. This methodological rigor resulted in significant findings concerning size variables, albeit with variations across different measures (assets, asset clusters, turnover, and sales by sales clusters). Consequently, a balanced panel of 1452 companies emerged, evenly split between innovative and non-innovative startups. The matching outcomes for each year are detailed in Table 3.

Subsequently, an in-depth annual analysis was performed using *t*-tests of means and DID; detailed results of these analyses are available upon request but are not presented here for brevity. We combined the post-match yearly data into a pooled dataset to construct an aggregate view, mitigating any potential year-specific biases. This compilation enabled the calculation of a 5-year survival rate, contrasting innovative startups with their non-innovative counterparts. The post-matching data description, including geographic distribution and incorporation form, reveals an equitable distribution between innovative and non-innovative firms, reflecting the criteria employed for matching. The same parity is observed in technology intensity as categorized by the NACE code, as delineated in Table 4.

A preliminary result of our analysis is the survival rate of the two panels: it is higher in the innovative startups than in the non-innovative ones, according to multiple time horizons (Table 5-A). This could signal the positive effect of the

TABLE 2 Distribution of Italian startups by region, year, legal form, and technology intensity before matching.

Panel A		Distribution by region and year					
		North W.	North. E.	Centre	South	Insular	Total
2014							
Innovative		11	2	4	12	0	29
Non-innovative		3396	2704	3588	3562	1287	14,537
Total		3407	2706	3592	3574	1287	14,566
2015							
Innovative		51	50	45	23	7	176
Non-innovative		1629	1413	1540	1392	287	6261
Total		1680	1463	1585	1415	294	6437
2016							
Innovative		93	96	52	65	10	316
Non-innovative		4476	3520	4687	4913	1765	19,361
Total		4569	3616	4739	4978	1775	19,677
2017							
Innovative		112	148	60	46	16	382
Non-innovative		2726	2201	2470	2271	772	10,440
Total		2838	2349	2530	2317	788	10,822
Grand total							51,502
Panel B		Distribution by legal form and year				Total	
		Joint-stock company	Limited liability company	Simplified limited liability company	Cooperative limited liability company		
2014							
Innovative	0	26	3	0		29	
Non-innovative	96	10,121	4320	0		14,537	
Total	96	10,147	4323	0		14,566	
2015							
Innovative	3	164	9	0		176	
Non-innovative	41	4199	2021	0		6261	
Total	44	4363	2030	0		6437	
2016							
Innovative	3	287	26			316	
Non-innovative	76	11,282	8003			19,361	
Total	79	11,569	8029	0		19,677	
2017							
Innovative	4	354	21	3		382	
Non-innovative	46	6557	3788	49		10,440	
Total	50	6911	3809	52		10,822	
Grand total							51,502
Panel C		Distribution by technology intensity and year			Total		
		Low-tech	High-tech				
2014							
Innovative		14	15			29	
Non-innovative		13,099	1438			14,537	
Total		13,113	1453			14,566	

TABLE 2 (Continued)

Panel C	Distribution by technology intensity and year		Total
	Low-tech	High-tech	
2015			
Innovative	69	107	176
Non-innovative	5017	1244	6261
Total	5086	1351	6437
2016			
Innovative	136	180	316
Non-innovative	18,047	1314	19,361
Total	18,183	1494	19,677
2017			
Innovative	152	230	382
Non-innovative	9302	1138	10,440
Total	9454	1368	10,822
Grand total			51,502

Note: The Table shows the distribution of Italian startups by region, year, legal form, and technology intensity for the population (before matching). We refer to Innovative as startup firms in Italy's manufacturing sector, registered between 2014 and 2017. High-tech and low-tech (a dummy variable differentiating the two groups of firms according to technology intensity) according to the European Union's classification, using the NACE-rev. 2 code.

TABLE 3 Matching results.

Treatment = innovative				
	2017	2016	2015	2014
Variables	Sales	Asset	Asset_cluster	Earning before tax
Asset_cluster	−0.403*** −0.126			
Sales_cluster		−1.108*** (0.323)		
Sales			−0.00214*** (0.00038)	
Assets				−0.0028** (0.0011)
NACE	−0.00016*** (0.000)	−0.0002*** (0.000)	−0.000087*** (0.000)	−0.00009* (0.000)
Legal	−0.592*** (0.060)	−0.653*** (0.075)	−0.882*** (0.130)	−0.503** (0.214)
Region	−0.131*** (0.023)	−0.061*** (0.022)	−0.047 (0.033)	−0.015 (0.057)
Constant	1.387*** (0.256)	1.272*** (0.382)	0.584** (0.296)	1.185** (0.520)
Observations	7561	13,683	4623	9582
Untreated	7251	13,426	4486	9560
Treated	310	257	137	22
Pseudo R ²	.0897	.0889	.109	.0695

Note: The Propensity Score Matching is carried out by modeling the probability of participation and matching participants with non-participants based on this probability, utilizing the one-to-one nearest-neighbor technique. We conducted year-by-year matches, considering various characteristics: (i) NACE activity code, (ii) legal form, and (iii) geographical location. Sales represent the total turnover, Asset is the value of total assets, and the cluster group sales and assets according to the size (quartiles). Standard errors in parentheses, *** $p < .01$, ** $p < .05$, * $p < .1$.

TABLE 4 Distribution by region, legal form, and technology intensity after matching.

Panel A	Region										
	North W.		North E.		Center		South		Insular		Total
Innovative	189	26.0%	249	34.3%	137	18.9%	121	16.7%	30	4.1%	726
Non-innovative	196	27.0%	237	32.6%	131	18.0%	125	17.2%	37	5.1%	726
Total	385	26.5%	486	33.5%	268	18.5%	246	16.9%	67	4.6%	1452
Panel B	Incorporation form										
	Joint-stock company		Limited liability company		Simplified limited liability company		Cooperative limited liability company				Total
Innovative	8	1.1%	663	89.2%	52	7.0%	20	2.7%			743
Non-innovative	8	1.1%	662	89.1%	51	6.9%	22	3.0%			743
Total	16	1.1%	1325	89.2%	103	6.9%	42	2.8%			1486
Panel C	Industry classification by technology intensity										
	Low-tech				High-tech						Total
Innovative	299		41.2%	427	58.8%						726
Non-innovative	402		55.4%	324	44.6%						726
Total	701		48.3%	751	51.7%						1452

Note: The Table shows the distribution of Italian startups by region, legal form, and technology intensity after matching. We refer to Innovative as startup firms in Italy's manufacturing sector, registered between 2014 and 2017. High-tech and low-tech (a dummy variable differentiating the two groups of firms according to technology intensity) according to the European Union's classification, using the NACE-rev. 2 code.

TABLE 5 Survival by year and region.

	Survival ^a																	
Panel A	1year		2years		Survival		3years		Survival		4years		Survival		5years		Survival	
Innovative	726		724		99.7%		721		99.3%		703		96.8%		684		94.2%	
Non-innovative	726		707		97.4%		669		92.1%		639		88.0%		609		83.9%	
Total	1452		1431		98.6%		1390		95.7%		1342		92.4%		1293		89.0%	
Panel B	Survival at 5years by region																	
	North W.			North E.			Center			South			Insular			Total		
Active	339	88.1%		430	88.5%		244	91.0%		220	89.4%		60	89.6%		1293	89.0%	
Inactive	46	11.9%		56	11.5%		24	9.0%		26	10.6%		7	10.4%		159	11.0%	
Total	385	100%		486	100%		268	100%		246	100%		67	100%		1452	100%	

Note: The Table presents the survival analysis by year and region. Panel A shows the single-year and cumulative survival rates of innovative and non-innovative startups. We refer to Innovative as startup firms in Italy's manufacturing sector, registered between 2014 and 2017. Whereas Panel B differentiates the 5-year survival rate between Active and Inactive startups.

^aBased on Bureau van Dijk data.

framework set up by the industrial policy to promote innovative startups. Differences among geographical areas are irrelevant, with a slightly higher survival rate in the central regions (Table 5-B).

Furthermore, due to our interest in innovativeness as measured by the patenting power, we also provide a sample description based on innovation activity, firstly by the patents already granted, as shown in Table 6. We can notice that the number of cases of companies that have obtained the recognition of patents is very low and located mainly in the North.

The patent process is a long one. Therefore, it is appealing to consider the propensity to innovate, fill in the documents for publication, and analyze the number of pending patents, which, in this case, is almost triple. A better picture is the

TABLE 6 Patent distribution.

Companies with granted patents									
Innovative	56		7.7%		726				
Non-innovative	23		3.2%		726				
Total	79		5.4%		1452				
	North W.	North E.	Center	South	Insular	Total			
With patents	28	32	6.6%	4.5%	6	1	1.5%	79	5.4%
No patents	357	455	92.7%	93.4%	249	66	97.6%	1383	94.6%
Total	385	487	100%	100%	255	67	100%	1462	100%
Companies with patents (granted and pending) by region									
	North W.	North E.	Center	South	Insular	Total			
With patents	64	93	16.6%	19.1%	32	7	10.4%	218	15.0%
No patents	321	393	83.4%	80.9%	236	60	91.1%	1234	85.0%
Total	385	486	100%	100%	268	67	100%	1452	100%
Companies with patent intensity by region									
	North W.	North E.	Center	South	Insular	Total			
Moderate	42	67	65.6%	72.0%	23	5	90.9%	157	72.0%
Strong	22	26	34.4%	28.0%	9	2	9.1%	61	28.0%
Total	64	93	100%	100%	32	7	100%	218	100%

Note: The Table demonstrates the innovation activity measured by Patents (granted and pending) and intensity. We refer to Innovative as startup firms in Italy's manufacturing sector, registered between 2014 and 2017. Patent intensity is measured as Moderate (1–5 patents) and Strong (With more than 5 patents).

intensity of patent publication by company, using the cut-off of 5 patents: most are moderate innovators, and strong innovators are mainly located in the northwest (Tables 7 and 8).

4.3 | The models

To answer the research question regarding the Startup Act policy evaluation, “Do innovative startups outperform their traditional peers?” We use the following models:

The *t*-test of means is employed in this study as a robust statistical tool to assess whether significant differences exist between the treated and untreated groups across key variables. Specifically, the *t*-test is appropriate in this context as it facilitates a straightforward and statistically rigorous comparison of the average values of financial indicators (e.g., financial size, profitability, financial structure, and growth) and innovation metrics between innovative and non-innovative firms. This method is advantageous because of the following features: Group Comparisons: The *t*-test directly evaluates whether the observed differences in means between the two groups—treated (innovative firms) and untreated (non-innovative firms)—are statistically significant, offering clear insights into the impact of innovation status on financial and innovation-related indicators. Simplicity and Interpretability: As a parametric test, the *t*-test provides a quick and interpretable way to analyze differences in means, assuming the data satisfies conditions such as approximate normality and homogeneity of variances. Given the variables under examination, these assumptions are reasonable and ensure the validity of the test. Relevance to research questions: The *t*-test aligns directly with the study's objectives, which include determining whether innovative firms differ systematically from non-innovative ones. This method can validate or reject the hypothesis that innovation is associated with measurable differences in financial performance and other relevant indicators. Targeted Insights into Group Dynamics: Beyond offering a “quick overview,” the *t*-test uncovers nuanced group dynamics, such as whether innovation leads to statistically significant improvements (or reductions) in specific financial metrics, providing actionable insights into the link between innovation and economic performance.

By using the *t*-test, we ensure a methodologically sound approach to exploring mean differences, bolstering the study's reliability and the interpretability of its findings. Furthermore, we have checked for the necessary assumptions of the *t*-test (e.g., normality, equality of variances) and addressed any deviations using appropriate corrections, such as Welch's *t*-test, when variances are unequal. The *t*-value and degrees of freedom for a paired *t*-test are calculated as follows:

$$T = \frac{\text{mean1} - \text{mean2}}{\frac{s(\text{diff})}{\sqrt{n}}} \quad (2)$$

where *mean1* and *mean2*=The average values of treated and untreated groups matched with Propensity Score Matching; *s(diff)*=The standard deviation of the differences of the paired data values; *n*=The sample size (the number of paired differences); *n* − 1=The degrees of freedom.

The panel probit regression model estimates the likelihood of a firm being classified as an innovative startup versus non-innovative based on financial, innovation, time-varying factors, and firm-fixed characteristics like region and legal form. This method is appropriate as the dependent variable is binary, and probit regression, with its cumulative normal distribution function, provides robust probability estimates.

The panel data structure allows the model to capture both cross-sectional and temporal dynamics, accommodating firm-level fixed effects that control for unobserved heterogeneity, such as regional and legal differences. This is essential to isolate the impact of financial and innovation activity variables, which are assumed to drive the decision to register as innovative.

Additionally, the probit model handles non-linear relationships between predictors and innovation probability, providing actionable insights into how financial and innovation factors interact with firm-specific traits to influence innovation status, ensuring methodological rigor and relevance. To avoid confounding and causality, independent variables at (*t* − 1) are assumed to lead the innovation registry at time *t*.

$$p(\text{innov})_{it} = \alpha + \beta_1 \text{FINN}_{i,t-1} + \beta_2 \text{INNOV}_{i,t-1} + \gamma_1 D_{\text{hitech}} + \gamma_2 D_{\text{region}} + \gamma_3 D_{\text{year}} + \varepsilon_t \quad (3)$$

where *p(innov)* is the probability of the startup to register as “innovative”, *FINN* is a set of financial performance variables, *INNOV* is a set of innovation proxies, *hitech* is a dummy for technology intensity (a dummy variable that differentiates the two

TABLE 7 Difference in means in financial variables.

	Innovative		Non-innovative			
	Obs	Mean	Obs	Mean	Diff.	Prob($T > t$)
Size						
<i>Sales_1</i>	726	50.10	726	179.48	−129.38	***
<i>Sales_2</i>	711	200.31	681	818.87	−618.55	***
<i>Sales_3</i>	693	310.22	619	1252.74	−1252.74	***
<i>Sales_4</i>	657	375.16	580	1411.23	−1036.07	***
<i>Sales_5</i>	609	510.51	529	1625.77	−1115.26	***
<i>Asset_1</i>	726	183.02	726	324.13	−141.11	***
<i>Asset_2</i>	711	394.97	681	792.65	−397.68	***
<i>Asset_3</i>	693	562.89	619	1065.39	−502.50	**
<i>Asset_4</i>	657	749.33	580	1329.92	−580.59	**
<i>Asset_5</i>	609	934.63	529	1641.45	−706.82	**
<i>Equity_1</i>	726	57.82	726	61.27	−3.45	
<i>Equity_2</i>	711	81.22	681	247.43	−166.22	*
<i>Equity_3</i>	693	126.66	619	355.89	−229.23	**
<i>Equity_4</i>	657	200.56	580	440.16	−239.60	**
<i>Equity_5</i>	609	275.45	529	548.79	−273.34	**
Profitability						
<i>ROE_1^a</i>	634	−0.41	662	−0.27	−0.14	
<i>ROE_2^a</i>	588	−0.38	586	−0.10	−0.29	**
<i>ROE_3^a</i>	561	−0.61	512	−0.07	−0.55	***
<i>ROE_4^a</i>	550	−0.87	508	−0.01	−0.86	***
<i>ROE_5^a</i>	508	−0.84	469	−0.11	−0.73	
<i>ROE_avg^a</i>	376	−0.40	386	0.02	−0.42	**
<i>ROA_1</i>	726	−0.15	725	−0.09	−0.06	
<i>ROA_2</i>	711	−0.12	681	−0.22	0.11	
<i>ROA_3</i>	693	−0.12	619	−98.34	98.22	
<i>ROA_4</i>	657	−0.12	580	−0.05	−0.08	*
<i>ROA_5</i>	609	−0.19	529	−0.06	−0.13	*
<i>ROA_avg</i>	596	−0.12	520	−0.02	−0.10	***
Financial structure						
<i>Lev_1^a</i>	639	0.47	664	0.37	0.10	***
<i>Lev_2^a</i>	590	0.33	588	0.29	0.04	***
<i>Lev_3^a</i>	563	0.31	514	0.28	0.03	***
<i>Lev_4^a</i>	554	0.32	510	0.29	0.03	*
<i>Lev_5^a</i>	512	0.33	471	0.29	0.04	**
<i>Lev_avg^a</i>	381	0.38	391	0.31	0.07	***
<i>Loan_1</i>	726	0.04	725	0.34	0.01	
<i>Loan_2</i>	711	0.09	681	0.05	0.04	***
<i>Loan_3</i>	693	0.10	618	0.06	0.04	***
<i>Loan_4</i>	657	0.12	580	0.06	0.06	***
<i>Loan_5</i>	609	0.12	529	0.08	0.04	***
<i>Loan_avg</i>	605	0.06	524	0.03	0.02	***

(Continues)

TABLE 7 (Continued)

	Innovative		Non-innovative		Diff.	Prob($T > t$)
	Obs	Mean	Obs	Mean		
Growth						
$\Delta Sales$	534	72.62	482	176.51	-103.88	
$\Delta Asset$	609	59.53	528	85.78	-26.25	

Note: The Table presents initial evidence of significant differences between innovative and their non-innovative counterparts in terms of financial performance and survival. We refer to Innovative as startup firms in Italy's manufacturing sector, registered between 2014 and 2017. Variables are measured for 5 years from the registration year to track the startup's survival. Sales is the total turnover from year 1 to year 5 from the registration. Assets represent the total assets, and Equity is the total equity. ROE is the return on equity for 5 years after the registration, ROE_avg is the 5-year average. ROA is the return on assets for 5 years after the registration, ROA_avg is the average ROA of the period. Lev is the firm's leverage for 5 years after the registration, measured as debt-to-assets, Lev_avg is the average period leverage. Loan is the percentage of bank loans to assets, Loan_avg is the percentage of the period bank loans. And $\Delta Sales$ and $\Delta Asset$ are the 5-year growth in sales and assets of the non-negative equity startups. *** $p < .01$ ** $p < .05$ * $p < .1$.

^aOnly for firms with non-negative equity.

TABLE 8 Difference in means in innovativeness measures.

Innovativeness	Innovative		Non-innovative		Diff.	Prob($T > t$)
	Obs	Mean	Obs	Mean		
Cum_grant_pat	726	0.08	726	0.03	0.04	***
Strong_pat	726	0.07	726	0.02	0.05	***
Moderate_pat	726	0.17	726	0.05	0.12	***

Note: The Table presents initial evidence of significant differences between innovative and non-innovative counterparts in terms of innovation performance. We refer to Innovative as startup firms in Italy's manufacturing sector, registered between 2014 and 2017. Cum_grant_pat (cumulative granted patents). Strong_pat (strong patent with more than 5 patents). Moderate_pat (moderate patent with less than 5 patents). *** $p < .01$ ** $p < .05$ * $p < .1$.

groups of firms; hitech and low-tech), whereas the remaining is region and registration year fixed effects. The following diagnostic checks (Goodness-of-Fit with Pseudo R-squared, Specification Test, and Influence of Outliers with Cook's Distance) are applied to ensure robustness and reliability.

4.3.1 | Panel regression for legal startup requirements

Panel regression models analyze the relationship between legal requirements for qualifying as an innovative startup and their impact on financial performance, innovative outcomes, and firm survival. This method is appropriate for the following reasons: (1) Continuous Outcomes: Panel analysis effectively handles continuous dependent variables, such as financial and innovative performance metrics. (2) Panel Structure: By leveraging the panel data, the model accounts for both time-varying and firm-specific effects, capturing the dynamic relationship between requirements and performance. (3) Focused Sample: Since only innovative startups are required to report these criteria, Panel analysis ensures that the analysis is both targeted and relevant to the specific regulatory framework. (4) Actionable Insights: This approach identifies how legal requirements shape multidimensional performance, offering clarity for both policymakers and firms. This ensures a rigorous and precise evaluation of the effects of startup qualification criteria. The model is specified as follows:

$$y_{it} = \alpha + \beta_1 REQ_{i,t} + \gamma_1 D_{hitech} + \gamma_2 D_{region} + \varepsilon_t \quad (4)$$

where y is a set of dependent variables, including financial, innovation, and the survival variable estimated separately, REQ is a set of binding requirements for firms to qualify as innovative startups. It is a binary variable showing whether the firm fulfills any of the legal requirements separately and collectively (R&D percentage, number of patents, and personnel with Ph.D.). The rest of the variables are the same as above.

In this regard, we recognize the importance of properly accounting for the panel structure when employing OLS regression. In our analysis, we utilized a fixed-effects model to address unobserved heterogeneity across entities, and we applied robust standard errors to mitigate issues of autocorrelation and heteroskedasticity.

5 | RESULTS

Differences-in-Differences (DiD) analysis reveals that innovative startups tend to be smaller across various economic indicators, including turnover, investments, assets, and equity. Profitability trajectories for startups reveal a temporal lag before achieving positive financial outcomes. A prevalent trend among many is the generation of losses, leading to instances of negative equity. This phenomenon is underscored by observing contrasting Return on Equity (ROE) with Return on Assets (ROA), illustrating that, on average, innovative firms accrue more substantial losses due to their investments in research and development (R&D) and innovation activities.

In terms of financing, innovative startups rely more on equity than their non-innovative counterparts, which is likely a beneficial outcome of industrial policies that offer tax incentives for such fundraising activities. Additionally, these startups secure a higher proportion of their debt financing from banks, potentially facilitated by public guarantees on bank loans. This financing structure suggests a more robust financial structure for innovative startups.

When considering growth metrics such as turnover and total assets, non-innovative firms appear to outpace their innovative peers, although the disparities are not statistically significant. A critical aspect of innovativeness, as indicated by the portfolio of patents within the sample, presents a noteworthy distinction: the count of innovative firms possessing at least one patent triples that of non-innovative firms, emphasizing the innovative essence of companies registered as such.

However, firms lacking innovative classification through patents demonstrate a more market-oriented approach, evidenced by higher scores in total quality, market attractiveness, and assignee score—indicative of a strategic re-inventiveness to safeguard technology and prevent market penetration by competitors. Moreover, legal scores, reflecting technology's protection and strategic positioning, also favor these firms. Despite these advantages, innovative startups lead in terms of intellectual property (IP) relevance, a metric signifying the centrality of IP to the firm's operations and its significant impact on financial performance. Nonetheless, when assessing IP efficiency—measuring the adaptability and early innovativeness of new technologies—the difference between innovative and non-innovative startups remains minimal and statistically irrelevant.

5.1 | Probit analysis

In Table 9, a probit analysis was conducted to ascertain the attributes influencing the propensity to register as an innovative startup. The probit model results (Table 9) reveal that financial leverage (coefficient ≈ 1.10 – 1.28) and bank loan ratios (coefficient ≈ 3.18 – 3.25) are significantly and positively associated with the probability of a firm registering as an innovative startup. Economically, this suggests that firms with stronger access to external financing—particularly debt—are more likely to engage with the innovative startup regime, perhaps due to the capital-intensive nature of innovation and the incentives available under the policy. The relatively large coefficient on loan ratios indicates that bank financing is pivotal in facilitating innovation-related activities in early-stage firms.

Beyond serving as a distinguishing characteristic between innovative and non-innovative startups, intellectual property (IP)—especially patents—plays a deeper strategic and financial role in the trajectory of firm development. Our findings indicate that startups meeting IP-related statutory requirements demonstrate higher survival rates and build more robust patent portfolios over time. To unpack this further, it is essential to consider how IP functions as both a signaling mechanism and a financial asset.

Patents help mitigate financial constraints by reducing information asymmetries between startups and external financiers. In early-stage ventures, where tangible assets are limited and performance histories are short, patent ownership can signal innovation potential and technological competency. This signal lowers perceived investment risk, enabling better access to bank financing and equity capital—particularly salient in financial ecosystems like Italy's, where venture capital remains underdeveloped, and banking institutions exhibit conservative lending practices.

Furthermore, IP enhances survival prospects by offering firms a legal and strategic moat around their innovations. Patents can help secure market share, attract strategic alliances, and open opportunities for licensing revenue. Startups with protected innovations are better positioned to withstand competitive pressures and to navigate the longer gestation periods often required before reaching profitability. This dual role of IP—as both a financing facilitator and a mechanism for sustaining competitive advantage—explains its robust association with the survival of innovative startups in our analysis. Thus, IP is not merely a correlate of innovativeness; it is a structural enabler of financial access and long-term viability, particularly in environments marked by capital scarcity and high startup mortality.

TABLE 9 Probit analysis for the likelihood of filing an innovative startup.

Variables	(1) Innov	(2) Innov	(3) Innov	(4) Innov	(5) Innov
Loans	3.230*** (0.696)	3.175*** (0.704)	3.254*** (0.710)	3.272*** (0.712)	3.183*** (0.731)
Leverage	1.277*** (0.220)	1.198*** (0.222)	1.110*** (0.225)	1.121*** (0.225)	1.102*** (0.233)
ROE	−0.185*** (0.0569)	−0.173*** (0.0577)	−0.190*** (0.0583)	−0.186*** (0.0584)	−0.351*** (0.0839)
Assets growth	−0.000127 (8.72e-05)	−0.000152* (8.67e-05)	−0.000161* (8.59e-05)	−0.000173** (8.81e-05)	−0.000187** (9.08e-05)
Strong_patent		0.747*** (0.246)	0.673*** (0.248)	0.639** (0.249)	0.561** (0.256)
Low_Tech			−0.380*** (0.0963)	−0.395*** (0.0972)	−0.398*** (0.101)
2015.Reg_year				0.125 (0.304)	0.131 (0.315)
2016.Reg_year				−0.0636 (0.294)	−0.103 (0.304)
2017.Reg_year				−0.0406 (0.293)	−0.0403 (0.303)
Region 2					−0.0946 (0.131)
Region 3					0.139 (0.140)
Region 4					−0.0750 (0.146)
Region 5					−0.118 (0.234)
Constant	−0.589*** (0.0985)	−0.589*** (0.0987)	−0.385*** (0.112)	−0.362 (0.299)	−0.253 (0.320)
Pseudo R ²	.0577	.0674	.0824	.0844	.0946
Observations	753	753	753	753	713

Note: The Table shows the results of the Probit analysis for the likelihood of filing an innovative startup. Innov (a dummy variable for being innovative; 1 innovative and 0 otherwise). Strong patent (more than 5 patents). Loan is the percentage of bank loans to assets. Lev is the firm's leverage measured as debt-to-assets. ROE is the return on equity. Asset growth is the percentage change in assets. Low_Tech (a dummy for low-tech startups). Reg_year (is the year of registration as an innovative startup). Region (is the geographical region). Standard errors in parentheses, *** $p < .01$, ** $p < .05$, * $p < .1$.

Conversely, the negative coefficients on Return on Equity (ROE) suggest that less profitable firms are more likely to seek innovative classification. This may imply that startups facing short-term profitability challenges leverage policy incentives (e.g., tax relief, public guarantees) to sustain their innovation investments.

Expanding the analytical framework, Model 2 incorporates an additional predictor related to the firm's innovation capacity, specifically the presence of robust Patent registrations. The analysis indicates that startups with significant patent portfolios are more predisposed to apply for innovative startup classification. The strong patent variable shows a positive and significant effect (coefficient ≈ 0.56 – 0.75), indicating that firms with robust IP portfolios are more likely to be formally recognized as innovative. This aligns with the role of IP as both a signal of technological capability and a strategic asset that enhances funding opportunities and survival.

Further refinement in Model 3 introduces the variable of technological intensity, differentiating between low-tech and high-tech sectors based on the NACE classification. The findings suggest a diminished likelihood for low-tech firms

TABLE 10 Legal requirements and performance measures.

Variables	(1) Asset growth	(2) ROE	(3) Loans	(4) Leverage	(5) Cum_pat	(6) Strong_pat	(7) Surv_5
Req_R&D	26.84 (39.65)	0.0727 (0.548)	0.00451 (0.0133)	−0.00436 (0.0339)	−0.0378 (0.0476)	0.0337 (0.0293)	0.0143 (0.0274)
Req_HR	−19.12 (72.80)	0.385 (1.197)	−0.00132 (0.0218)	0.0683 (0.0740)	−0.0615 (0.0838)	−0.0455 (0.0516)	0.0262 (0.0483)
Req_IP	6.868 (38.35)	0.148 (0.529)	0.0200 (0.0128)	−0.00599 (0.0326)	0.209*** (0.0458)	0.0990*** (0.0281)	0.0463* (0.0264)
Req_cum	−0.842 (81.69)	0.171 (1.292)	−0.0196 (0.0252)	−0.0143 (0.0798)	0.0380 (0.0950)	0.0618 (0.0584)	−0.0533 (0.0547)
High_Tech	−5.915 (25.89)	0.363 (0.387)	−0.00965 (0.00852)	0.0693*** (0.0239)	0.0977*** (0.0305)	0.0303 (0.0188)	−0.00883 (0.0176)
Region 2	48.55 (33.51)	−0.236 (0.502)	−0.00404 (0.0109)	−0.00924 (0.0307)	−0.0123 (0.0392)	−0.0150 (0.0241)	0.0286 (0.0226)
Region 3	9.620 (36.37)	0.166 (0.526)	−0.0157 (0.0120)	−0.0234 (0.0326)	−0.0759* (0.0434)	−0.0474* (0.0267)	0.0407 (0.0250)
Region 4	5.843 (38.02)	−1.032* (0.555)	0.00129 (0.0126)	0.000838 (0.0340)	−0.0353 (0.0449)	−0.00904 (0.0276)	−0.0185 (0.0259)
Region 5	−15.59 (63.82)	0.0894 (0.937)	0.0292 (0.0214)	−0.120** (0.0579)	−0.0704 (0.0772)	−0.0233 (0.0475)	0.00717 (0.0445)
2015.Reg_year	−199.1*** (75.20)	−0.127 (1.281)	0.0150 (0.0247)	−0.0626 (0.0728)	0.171* (0.0931)	0.107* (0.0573)	0.0619 (0.0537)
2016.Reg_year	−285.5*** (73.27)	−0.535 (1.258)	0.0108 (0.0241)	−0.0180 (0.0712)	0.141 (0.0903)	0.0673 (0.0555)	0.0315 (0.0520)
2017.Reg_year	−301.4*** (73.42)	−0.0225 (1.260)	−0.000955 (0.0240)	−0.0105 (0.0713)	0.0495 (0.0898)	0.00879 (0.0552)	−0.000146 (0.0517)
Constant	329.0*** (82.86)	−0.417 (1.381)	0.0520* (0.0273)	0.362*** (0.0791)	0.0442 (0.102)	−0.0396 (0.0626)	0.890*** (0.0587)
Observations	579	376	605	381	726	726	726
R-squared	.048	.024	.032	.053	.122	.064	.033

Note: The Table shows the impact of legal requirements on performance measurements and the survival of innovative startups. Asset growth is the percentage change in assets. ROE is the return on equity. Loan is the percentage of bank loans to assets. Lev is the firm's leverage measured as debt-to-assets. Cum_pat (cumulative granted patents). Strong patent (more than 5 patents). Surv_5 is the 5-year survival rate. Reg_year (is the year of registration as an innovative startup). Region (is the geographical region). Req_R&D (R&D requirement). Req_HR (Personnel requirement). Req_IP (Intellectual Property Requirements). Req_cum (All requirements together). High_Tech (a dummy for high-tech startups). Standard errors in parentheses. *** $p < .01$, ** $p < .05$, * $p < .1$.

to register as innovative startups compared to their high-tech counterparts, underscoring the role of technological sophistication in this designation. The negative coefficient on the low-tech dummy (≈ -0.38 to -0.40) reinforces that technological intensity is a key differentiator in innovation policy participation. Due to their R&D orientation and greater innovation potential, high-tech firms are more inclined—and better positioned—to meet the eligibility criteria of the Startup Act.

Subsequent analyses in Models 4 and 5 account for fixed characteristics, including regional location and the year of establishment. The results from these models suggest that regional disparities do not significantly influence the decision to register as an innovative startup, nor does the year of formation bear a statistically significant effect on this probability. This comprehensive examination underscores the multifaceted nature of factors contributing to the designation as an innovative startup, highlighting the importance of financial leverage, patent strength, and technological intensity while suggesting a relative uniformity across regions and establishment periods in the propensity to seek innovative classification.

These coefficients suggest that access to finance, IP strength, and technological positioning are statistical predictors of innovation classification and economically meaningful levers for policy targeting and firm growth strategy.

5.2 | Legal requirements analysis

Table 10 examines how innovative startups' legal requirements influence various performance metrics, including growth, financial health, leverage, innovation capacity, and survival. These requirements are categorized as follows: Requirement 1 (rd) for startups fulfilling the research and development criterion, Requirement 2 (hr) for those with personnel holding Ph.D. qualifications, Requirement 3 (intel_prop) for entities with patent holdings, and a cumulative requirement (req_cum) for firms meeting all three specified criteria.

The analysis reveals that these legal stipulations generally exert a negligible effect on the comprehensive performance metrics of the startups, with limited exceptions. Notably, entities demonstrating robust intellectual property activities, as evidenced by patent holdings, exhibit a higher propensity for survival over five years. Additionally, these firms are more adept at developing and augmenting their patent portfolios. Corroborating earlier findings, startups within the high-technology sector are characterized by increased leverage and a higher volume of patent accumulations. Distinctions in the impact of legal requirements are also observed across different geographical regions and the years of establishment, indicating variability in compliance and performance outcomes based on regional and temporal contexts.

6 | CONCLUSIONS

This study critically evaluates innovative startups' performance, financial structure, and survival dynamics under the Italian Startup Act (2012), set against their traditional counterparts. Utilizing robust matching techniques and analytical frameworks, this research uncovers critical insights into the policy's implications and the challenges confronting innovative startups in Italy.

Our findings illustrate that innovative startups, while requiring longer to achieve profitability, demonstrate superior survival rates and stronger financial structures, underscoring the policy's efficacy in mitigating financial barriers. These startups show an increased capacity to secure equity and leverage public loan guarantees, reflecting an evolved financing ecosystem. However, their high dependency on R&D investments often results in delayed profitability and financial fragility in their early stages.

The patenting activity serves as a distinguishing feature of innovative startups, with higher proportions of firms in this category obtaining patents compared to their non-innovative peers. Yet, the limited scale of patenting across the ecosystem highlights the underutilized potential for intellectual property development. Furthermore, stringent IP compliance correlates positively with enhanced survival rates and robust patent portfolios, emphasizing the critical role of intellectual property policies.

Geographical and temporal factors appear to exert minimal influence on the observed outcomes, indicating consistency in the policy's application across regions and registration periods. However, regional disparities in startup density and innovation intensity suggest the need for more localized support mechanisms to foster equitable growth.

Our findings are consistent with earlier studies that recognize the positive effects of the Italian Startup Act on alleviating credit constraints and enhancing survival prospects (Brancalani et al., 2022; Giraudo et al., 2019). However, our results diverge from Finaldi Russo et al. (2016), who found limited differences between innovative and non-innovative startups, particularly in the early years of the policy. By extending the time frame of analysis to five years and focusing specifically on the manufacturing sector, our study provides more nuanced evidence of long-term financial resilience and innovation outcomes. Moreover, we contribute novel insights by examining the interaction between legal requirements—especially intellectual property compliance—and performance indicators such as survival and patent activity. In doing so, we fill a critical gap in the literature that has largely overlooked the structural role of IP as both a signal of quality and a tool for navigating financial constraints. This comprehensive approach enables a deeper understanding of how policy design can more effectively support high-potential startups and foster innovation-driven growth.

6.1 | Policy implications and future research

The Italian Startup Act holds considerable potential as a catalyst for innovation and entrepreneurship, yet its full capabilities remain untapped. As Italy grapples with persistent financial volatility and an underexploited intellectual property system, it becomes increasingly clear that structural reforms are essential to unlock the Act's transformative promise.

A first step toward optimizing the framework lies in strengthening the foundation for research and development. By offering more targeted grants or fiscal incentives, the government can help startups weather early-stage financial losses while laying the groundwork for sustained innovation. Simultaneously, intellectual property support must be reimagined by simplifying the bureaucratic hurdles of patent applications and providing accessible, subsidized legal and technical resources that empower entrepreneurs to protect and leverage their innovations effectively.

However, national-level strategies alone will not suffice. Innovation ecosystems must be nurtured locally, with tailored policy interventions designed to bridge regional disparities and cultivate vibrant entrepreneurial hubs beyond Italy's traditional industrial centers. These localized efforts should be coupled with a strategic emphasis on high-tech sectors, statistically more inclined to generate scalable innovations. Policymakers can better support these high-potential ventures' unique dynamics and needs by crafting sector-specific incentives.

Finally, to ensure that these efforts translate into measurable progress, it is crucial to adopt a long-term lens. Ongoing monitoring and longitudinal studies should track startup development over time, assessing business performance and policy outcomes, and the broader societal effects of entrepreneurship. Such a comprehensive approach would refine existing policies and provide a roadmap for future innovation-led growth.

In sum, this study reinforces the critical role of targeted industrial policies like the Italian Startup Act in nurturing innovative ecosystems. However, continuous adaptation and refinement of these policies will be essential to sustain and expand their positive impacts on innovation, financial resilience, and economic dynamism.

Future research should expand the analytical timeframe, integrate comparative studies across European nations, and explore the systemic interplay between industrial policies and entrepreneurial ecosystems. Investigating the interplay between public and private financing and sectoral innovation strategies would yield actionable insights into fostering a more vibrant and inclusive startup landscape.

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Endnotes

¹ Art. 25 says "These provisions are aimed at promoting sustainable growth, technological development, new entrepreneurship and employment, especially among young people, with regard to innovative start-up companies".

² Red tape costs and the length of civil juridical trials are among the factors that limit the birth and the growth of firms (Calvino, et al., 2016).

³ Orbis is a commercial balance-sheet repository maintained by Bureau Van Dijk. For Italy it contains detailed accounting data for around four million companies.

⁴ Adopting the Eurostat (n.d) classification.

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